DECK 12 · WEEK 6 · MORNING 1

Equity II
Analysis & Valuation

From business model to intrinsic value: the five valuation tools every analyst owns

CFA Level I · Vol 5 Learning Modules 5–8

Roadmap — Equity II — Analysis & Valuation

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Company analysis — past & present

Research-report anatomy · business model · revenue drivers · pricing power · operating & financial leverage

02

Industry & competitive analysis

Classification (GICS) · Porter Five Forces · PESTLE · generic strategies · industry life cycle

03

Company forecasting

Four forecast approaches · top-down vs bottom-up revenue · scenario analysis (bear/base/bull) · iPad-vs-PC worked case

04

Valuation toolkit

DDM, Gordon & two-stage · FCFE · P/E, P/S, P/B · EV/EBITDA · asset-based — Siemens, IBM, Nestlé, Cameco

Learning objectives

By the end of this deck you should be able to:

1. Describe the structure and elements of an initial equity research report, and evaluate a company's business model, revenue drivers, and operating/financial leverage.

2. Analyze an industry's structure using Porter's Five Forces and identify the dominant forces to monitor, and complement the analysis with a PESTLE scan.

3. Apply the four forecast approaches to build a top-down or bottom-up revenue forecast and quantify key risks with bear / base / bull scenarios.

4. Calculate intrinsic value using the Gordon growth model, a two-stage DDM, and an FCFE model, and identify the company profile each is suited to.

5. Compute and interpret P/E, P/S, P/B, EV/EBITDA and asset-based valuations — and understand why a low multiple demands a "why" before a "buy".

SECTION 01

Company analysis past & present

An initial equity research report sits on three analytical pillars — company, industry, and forecast — all feeding valuation and a recommendation

Initial reports cover all eight elements below. Subsequent reports revisit only the ones that have changed — usually financials and valuation.

ANALYTICAL FRAMEWORK

3. FORECASTING
Revenue, profitability, working capital, capex, capital structure, risks

2. INDUSTRY & COMPETITIVE
Classification · size/growth/margins · Porter Five Forces · PESTLE · position

1. COMPANY — PAST & PRESENT
Business model · revenue drivers · operating margins · capital structure

INITIAL-REPORT ELEMENTS

1. Recommendation (buy / hold / sell) + target price
2. Company description & business model
3. Industry overview & competitive positioning
4. Financial analysis & model
5. Valuation (present-value + multiples)
6. ESG considerations
7. Ownership & management
8. Material upside / downside risks

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, Exhibits 1–2, pp. 198–204.

A company's business model answers four simple questions — nail these before touching the spreadsheet

Rigorous analysts verify answers with regulatory filings, investor events, site visits, and customer/supplier interviews — never management alone.

1

WHO BUYS?

Customer segments, geographic footprint, concentration. Is it one large customer (Apple → TSMC), many small ones (retail shopper), or a captive government buyer (defense)?

2

WHAT DO WE SELL?

Products & services, recurring vs one-off, discretionary vs staple, single-product vs multi-segment. Warehouse Club Inc. is single-segment; conglomerates are the other extreme.

3

HOW DO WE MAKE MONEY?

Unit economics: price × volume, margin structure, fixed vs variable cost split, working-capital cycle, required capital intensity. Drives operating leverage and returns on capital.

4

WHAT IS THE STRATEGY?

Cost leadership, differentiation or focus. Intentional strategy with measurement and feedback beats unintentional drift — and is usually only visible in hindsight.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, §3 "Business Model", pp. 204–220.

Revenue analysis decomposes growth into volumes, prices and mix — with pricing power the decisive lever

Pricing power is a function of industry structure (few concentrated suppliers, differentiation, switching costs) and strategy — not wishful thinking by management.

$$\text{Revenue} = \text{Volume} \times \text{Price} \qquad \Delta\text{Rev} \approx \%\Delta\text{Volume} + \%\Delta\text{Price} + \text{Mix} + \text{FX}$$

where: Mix = shift in product / geography weights; FX = currency translation for reporters in one functional currency.

CFA LM 5 SELF-ASSESSMENT Q4 — BEVERAGE PRICE × VOLUME WALK

Tea-based drink manufacturer: last year 2 m cases at USD 20 average price. This year: 3 m cases, ASP rises 5% (less discounting). Forecast net-sales growth?

1. Prior-year sales: USD 20 × 2 m = USD 40 m
2. Forecast ASP: USD 20 × 1.05 = USD 21
3. Forecast sales: USD 21 × 3 m = USD 63 m
4. Growth: 63 / 40 − 1 = +57.5%, rounded to 58%

Growth = +58%. Volume effect +50%, price effect +5%, interaction +2.5% — all three matter at once.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, self-assessment Q4 and §3 "Revenue Analysis", pp. 200, 204–212.

Operating and financial leverage multiply each other — total leverage is their product, and can make or break a cyclical business

Rule of thumb: a firm already running high operating leverage should be cautious with financial leverage — their product is what shareholders actually experience.

$$DOL = \%\Delta EBIT / \%\Delta Sales \qquad DFL = \%\Delta NI / \%\Delta EBIT \qquad DTL = DOL \times DFL$$

where: DOL = degree of operating leverage (from fixed costs). DFL = degree of financial leverage (from fixed interest). DTL = degree of total leverage.

CFA LM 5 SELF-ASSESSMENT Q6 — COMPANY XYZ DOL

Revenue 20X0: EUR 9,256 m → 20X1: EUR 9,707 m. Operating income: EUR 1,527 m → EUR 1,673 m. What is DOL for 20X1?

1. %ΔSales = 9,707 / 9,256 − 1 = +4.87%
2. %ΔEBIT = 1,673 / 1,527 − 1 = +9.56%
3. DOL = 9.56% / 4.87% = 1.96×

DOL = 1.96× — a 1% sales move becomes a 1.96% EBIT move. Any DFL on top amplifies the swing at the equity line.

Source: CFA Institute, Curriculum Vol 5, Learning Module 5, self-assessment Q6 & Q8 and §4 "Cost & Leverage", pp. 200–201, 221–245.

SECTION 02

Industry & competitive analysis

Industry structural factors set a long-run profitability ceiling — company-specific factors create the spread beneath it

Competition "pulls" company profitability to an industry base rate over time. Industry analysis estimates that base rate; company analysis estimates the gap to it.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
DEFINE	**SURVEY**	**STRUCTURE**	**EXTERNAL**	**POSITION**
Industry boundaries, constituents, substitutes	Size, growth, character, profitability, market share	Porter Five Forces — long-run profitability drivers	PESTLE — growth & share dynamics	Company strategy vs industry forces

WHY INDUSTRY MATTERS

Firms in the same industry face similar demand, factor costs, regulation, and substitutes. Structural forces (Porter) determine long-run returns on invested capital — the ceiling.

Company-specific factors — size, execution, strategy — determine how close a firm gets to that ceiling.

IMPROVED FORECASTS & IDEAS

Industry analysis sharpens forecasts by modelling the competitive forces that discipline prices, costs, and share.

It also surfaces investment candidates the bottom-up screen missed — the second-best in an attractive industry can beat the best in a structurally broken one.

[1] McGahan & Porter (1997) found industry effects matter most for the sustainability of economic profits; company-specific effects explain more variance for laggards than for leaders.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §2 "Uses of Industry Analysis", pp. 249–251. McGahan & Porter (1997).

GICS, ICB and TRBC are strict hierarchical taxonomies — useful starting points, but limited by multi-segment firms and geography

A "demand" grouping (similar products from the customer's perspective) has largely replaced legacy "supply" groupings like SIC/NACE/ISIC.

Scheme	Provider	Tiers (top → bottom)	# Sectors
GICS	MSCI + S&P Dow Jones	Sector → Industry Group → Industry → Sub-Industry	11 / 25 / 74 / 163
ICB	FTSE Russell	Industry → Supersector → Sector → Subsector	11 / 20 / 45 / 173
TRBC	Refinitiv	Econ. Sector → Business → Industry Group → …	14 / 33 / 62 / 154
SIC / NACE / ISIC	Government (US / EU / UN)	Legacy supply-based — updated infrequently	varies

FOUR LIMITATIONS ANALYSTS LIVE WITH

1. Overly narrow or broad groups (Shopify grouped with Check Point in "Application Software") · 2. Multi-product firms force a single bucket (Amazon in Consumer Discretionary despite AWS dominance) · 3. Global schemes miss local industries (healthcare services, regulated utilities) · 4. Groupings change — affects historical comparability (creation of Real Estate sector in 2016).

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §3 "Industry Classification", pp. 251–258.

Porter's Five Forces assesses long-run industry profitability — scored High / Moderate / Low on each of the five dimensions

Scoring is qualitative — if several forces are intense, almost no firm earns attractive returns on capital; if all are benign, the industry is a goldmine.

Force	US Retail score	Driver — why this rating?
Threat of new entrants	VERY HIGH	>40,000 new retail firm formations per month in the US; minimal licensing and patents
Threat of substitutes	LOW	Services (restaurants, travel) are only a partial substitute; most goods cannot be replaced by services
Bargaining power of customers	MODERATE	Fragmented consumers, but internet comparison-shopping drives price sensitivity on identical SKUs
Bargaining power of suppliers	LOW to MODERATE	Many manufacturers; branded goods with sole sources (luxury) exert more power than commodities

VERDICT

Retail is a high-rivalry, high-entry industry with moderate customer power — the median operator earns thin operating margins (~3–5%). Scale, scope, and brand loyalty separate leaders (Costco, Home Depot ~14%) from the pack. Amazon disrupted retail not by changing the forces but by changing the fundamental cost of serving a customer.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 Case Study "US Retail (ex autos)", pp. 270–271.

An analyst must rank the five forces — one or two usually dominate, and those are what must be monitored quarter to quarter

"What happens if company X raises prices by 10% tomorrow?" If customers switch easily or substitutes exist, buyer power and substitutes are the forces to watch.

BRANDED BEVERAGES

Dominant force: Supplier power

Concentrated bottler-distributor networks with exclusive shelf arrangements; few substitutes for brand equity. Coke and Pepsi extract the rents.

CLOUD INFRASTRUCTURE

Dominant force: Scale barriers

Massive fixed costs (server fleets, networks) and economies of scope with adjacent businesses deter new entrants. AWS / Azure / GCP lock in long-run margin.

GENERIC PHARMA

Dominant force: Rivalry

Chemically-equivalent products approved by regulators; pharmacies indifferent between suppliers. Price competition collapses margins to near marginal cost.

RETAIL BANKS (USA)

Dominant force: Entry barriers

<3 new commercial bank charters issued per year 2010–2021. Regulatory capital, deposit insurance, and licensing make de novo entry almost impossible.

COMMERCIAL AIRCRAFT

Dominant force: Rivalry

Duopoly (Airbus, Boeing) but fierce price / financing competition per sale. High exit barriers (specialized assets) keep weak players in.

BUSINESS SOFTWARE (ERP)

Dominant force: Switching costs

SAP / Oracle require years to implement and integrate with mission-critical systems. Customers are effectively locked in — incumbents price accordingly.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 "Industry Structure", pp. 265–271.

PESTLE catches what Porter misses — the external tides that move an industry's growth rate and share dynamics

Porter asks "how attractive is the industry?". PESTLE asks "what outside the industry is about to change?". Use both.

Factor	What it covers	Beauty-industry example
P — Political	Tax, regulation, geopolitical, trade, subsidies	Proposed packaging-tax on plastic containers
E — Economic	GDP, income, inflation, rates, FX	Emerging-market income growth expands premium-SKU demand
S — Social	Demographics, culture, lifestyle, consumer tastes	Selfie & social-media culture lifts premium-skincare demand
T — Technological	Innovation (sustaining vs disruptive)	AR "try-on" filters shift customer discovery to online
L — Legal	Contract, competition, IP, labor laws	Trademark enforcement on counterfeit cosmetics in cross-border e-commerce

POLITICAL, NOT COMPETITIVE

A legislature debating a plastic-packaging tax is a PESTLE signal — political. A competitor launching AR filters is a COMPETITIVE move — Porter. Different frameworks, different responses.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §5 "External Influences", pp. 271–276.

Three generic competitive strategies — cost leadership, differentiation, focus — explain most of what companies actually do

Evaluating a strategy has three tests: (1) does it defend against the five forces? (2) does it align with PESTLE? (3) does the firm have the resources and capabilities to execute?

COST LEADERSHIP

Lowest unit cost in the industry.

Means:
- Scale economies from fixed costs
- Favorable raw-material access
- Strict cost culture
- Aggressive pricing for volume
- Low-cost distribution
- Economies of scope

Exemplars:
Costco · Walmart · Ryanair · Southwest Airlines · Vanguard

DIFFERENTIATION

Uniqueness customers will pay for.

Means:
- Brand and advertising
- Proprietary channels
- Trademarks, patents, copyright
- Superior quality / features
- Customer-experience culture
- Premium pricing

Exemplars:
Apple · LVMH · Ferrari · Rolex · Nike · Disney

FOCUS

Narrow segment served exceptionally.

Means:
- Proximity to a specific customer
- Deep understanding of their needs
- May combine cost OR differentiation within that niche

Exemplars:
Ferrari (luxury auto) · Edward Jones (rural US brokerage) · Hermès (ultra-luxury leather) · Ducati (sport motorcycles)

[1] "Stuck in the middle" — neither a cost leader, nor differentiated, nor focused — is the worst competitive position; rare to sustain economic profits.

Source: CFA Institute, Curriculum Vol 5, Learning Module 6, §6 "Competitive Positioning", pp. 276–285. Porter (1980).

SECTION 03

Company forecasting

There are four forecast approaches — historical, base-rate convergence, management guidance, and analyst discretionary — each fits a different setting

In practice, approaches are combined — e.g., use guidance for next year, base-rate convergence for the long run, discretionary for a disruptive inflection.

Approach	Best-fit setting	Pitfall
Historical results	Mature, non-cyclical industries; non-material line items	Past conditions may not repeat
Base rates & convergence	Young firms scaling toward industry norms; maturing markets	Industry average may be shifting too
Management guidance	Strong track record of accuracy; management controls the variable	Macro-sensitive lines — no info advantage
Analyst discretionary	Cyclicals, disruptive change, unique business, no comparables	Dependent on analyst judgment quality

FORECAST OBJECTS — 4 LEVELS OF GRANULARITY

1. DRIVERS of financial-statement lines (# stores × sales / store) — highest explanatory value. · 2. INDIVIDUAL LINES directly — efficient when drivers unclear or line is immaterial. · 3. SUMMARY MEASURES (FCF, EPS, total assets) — efficient but opaque. · 4. AD-HOC objects (one-off legal, regulatory, weather) — require judgment.

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §2 "Forecast Objects, Principles, Approaches", pp. 288–295.

Top-down revenue forecasts pivot on market size and share; bottom-up forecasts build from volumes and prices — cross-checking both uncovers hidden assumptions

Both methods applied to the same company should converge within a narrow band — if they don't, one or more assumptions is wrong and worth finding.

TOP-DOWN (MARKET × SHARE)

Step 1 — forecast market size
· nominal GDP × industry / GDP ratio
· adjust for life-cycle (growth, mature, decline)

Step 2 — forecast company share
· stable, gaining, losing — and by how much

Step 3 — multiply
· Revenue = Market size × Share

Example (Warehouse Club):
- US retail (ex autos) grows 3.4% p.a.
- Share gain: +2 bps per year
- 10-y cumulative top-line driver: +38%

BOTTOM-UP (VOLUME × PRICE)

Step 1 — forecast capacity / units
· stores opened, subscribers added, planes in fleet

Step 2 — forecast unit economics
· sales per store, ARPU, yield per seat-mile

Step 3 — multiply & aggregate
· Revenue = Σ (units × unit price) by segment

Example (Warehouse Club):
- 4–6 new stores / year
- Same-store sales growth 3.8%
- ~450 k new members / year × annual fee
- Totals: +USD 8–10 bn / year

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §3 "Forecasting Revenues", Exhibit 2, pp. 295–300.

A three-scenario forecast — bear, base, bull — disciplines the analyst into quantifying the risks already identified in Porter and PESTLE

Each scenario should correspond to a coherent story — bear = e-commerce share loss accelerates, base = trend continues, bull = physical stores regain share post-pandemic.

BASE CASE — US CONSUMER-STAPLES RETAILER (FY+1)

Starting revenue USD 11,500 m. Real GDP +2.0%, inflation +2.5% → nominal GDP +4.5%. Industry tracks GDP. Company share is stable at 0.70% of US retail ex-autos. Same-store sales +3.5%, +5 new stores at USD 50m each.

BASE-CASE FY+1 REVENUE ≈ USD 12,100 m (+5.2% YoY)

Scenario	Story	GDP	SSS	FY+1 rev USD m
Bear (25%)	Accelerating e-commerce share loss; recession	+0.5%	−1.5%	11,610
Base (50%)	Trend continues: mild GDP growth, share stable	+4.5%	+3.5%	12,100
Bull (25%)	Physical-store renaissance + margin expansion	+5.5%	+6.0%	12,990
Probability-weighted FY+1 revenue				**12,200**

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, §6 "Scenario Analysis", pp. 318–320.

Scenario analysis of a disruptive threat — the iPad cannibalizing PC demand — sizes potential Microsoft revenue losses with explicit assumptions

At the start of 2012, one year after iPad launch — the question: what fraction of PC sales will tablets replace, and what does that cost Microsoft?

Row	FY11	FY12E	FY13E	FY14E	3-y CAGR
Pre-cannib. PC shipments (k)	350,903	360,000	380,000	400,000	+4.5%
Tablet shipments (k)	38,471	90,000	125,000	175,000	+65.7%
Cannib. factor (consumer / non-consumer)	30% / 10%	30% / 10%	30% / 10%	30% / 10%	–
PCs cannibalized (k)	11,205	25,560	34,750	47,250	–
MSFT revenue impact (USD m, ASP 85 cons / 155 non)	964	2,223	3,050	4,200	+63%

BASE-CASE TAKEAWAY

Under the base case (30% consumer / 10% non-consumer cannibalization), Microsoft loses ~USD 2.2 bn of PC-OS + Office revenue in FY12, growing to ~USD 4.2 bn by FY14. Bear case (40%/15%) extrapolates to USD 6 bn+; bull case (20%/5%) halves the damage. The analyst's recommendation depends on which scenario is priced in.

Source: CFA Institute, Curriculum Vol 5, Learning Module 7, Example 4 "Tablet Cannibalization", pp. 319–323.

SECTION 04

Valuation toolkit

Three families of valuation models — present-value, multiplier, and asset-based — together triangulate an intrinsic-value estimate

Price < value → undervalued; price > value → overvalued; price ≈ value → fairly valued — always interpreted with an analyst confidence band.

PRESENT VALUE (DCF)

Intrinsic value = PV of future cash flows to equity.

Models:
- Gordon growth DDM
- Multi-stage DDM
- FCFE model

Best for:
Mature, dividend-paying, predictable firms with forecastable cash flows.

Weakness:
Sensitive to r and g; terminal value often dominates.

MULTIPLES

Intrinsic value inferred from comparable companies.

Metrics:
- P/E, P/B, P/S, P/CF
- EV/EBITDA, EV/Sales, EV/Op. Income

Best for:
Cross-sectional comparison within industry; quick screen for over-/under-valuation.

Weakness:
Assumes peers correctly priced; accounting noise; cyclicality distorts denominators.

ASSET-BASED

Intrinsic value = Σ assets − Σ liabilities − preferred, at fair values.

Best for:
- Holding companies, REITs, closed-end funds
- Natural-resource and financial firms
- Liquidations, takeover floors

Weakness:
Ignores intangibles (brands, people, growth). Provides a floor, not a going-concern value.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §3 "Categories of Valuation Models", pp. 338–340.

The DDM prices a share as the PV of future dividends — FCFE applies the same idea to dividend-paying capacity instead of actual dividends

Mature dividend-payer with stable payout? DDM. Young firm, non-payer, or levered firm with large buybacks? FCFE.

$$V_0 = \Sigma D_t / (1 + r)^t \qquad FCFE = CFO - FCInv + \text{Net borrowing}$$

where: D_t = expected dividend in year t; r = required return on equity; FCInv = fixed-capital investment (capex); Net borrowing = new debt − repayments.

USE DDM WHEN

- The firm pays a stable, predictable dividend
- Dividends track earnings at a consistent payout
- Management has signaled dividend policy credibly
- Examples: mature utilities, consumer staples, tobacco

Pitfalls: very low or zero dividends force assumptions; buybacks distort the "dividend" proxy.

USE FCFE WHEN

- The firm doesn't pay dividends (Amazon, Meta pre-2022)
- Dividends are disconnected from capacity to pay
- Heavy buybacks supplement or replace dividends
- Leverage changes meaningfully affect equity CF

FCFE requires forecasting CFO, capex, working capital, and net borrowing — more inputs but less arbitrary.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §5 "DDM and FCFE Model", pp. 343–347.

The Gordon growth model prices a stock as a growing perpetuity — elegant, but extremely sensitive to r and g

Cross-reference Deck 01 Shipline — there we used D_0 = 2.00, r = 10%, g = 6%→3%. Here CFA uses Siemens with D_0 = 3.70, r = 7.5%, g = 5.4%.

$$V_0 = D_0 \times (1 + g) / (r - g) = D_1 / (r - g)$$

where: Assumes r > g (else infinite); constant g forever; constant r forever; dividends are the right cash-flow measure.

CFA LM 8 EXAMPLE 7 — SIEMENS AG (FY2017)

Most recent dividend D_0 = 3.70. Dividend growth 2013→2017 averaged 5.4%. Required return estimated at 7.5%.

1. D_1 = 3.70 × 1.054 = 3.90
2. r − g = 7.5% − 5.4% = 2.1%
3. V_0 = 3.90 / 0.021

V_0 = €185.70 vs market €119.20 → appears undervalued

r ↓ / g →	3.5%	4.5%	5.5%
7%	109	155	260
8%	85	110	156
9%	70	86	112
10%	59	70	87

[1] The numerator in the Gordon formula is D_1 (next-year dividend), not D_0. Using the wrong numerator is a common mistake.
[2] Sustainable growth check: g = b × ROE, with b = retention rate = 1 − payout. Siemens implied g ≈ 8.5% using 5-year average — higher than the dividend-growth estimate of 5.4%.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §7 "Gordon Growth Model", Example 7, pp. 349–352.

The two-stage DDM bolts a Gordon terminal onto an explicit high-growth phase — the workhorse for firms transitioning to maturity

IBM was a transitioning firm in 2018 — 5-year dividend growth 13.5%, but forecast 4.5% through 2023. Two-stage DDM designed for exactly this profile.

$$V_0 = \Sigma_{t=1..n} D_0(1+g_S)^t / (1+r)^t + [D_{n+1} / (r - g_L)] / (1+r)^n$$

where: g_S = short-term growth during stage 1 (n years); g_L = long-term sustainable growth; terminal value = Gordon perpetuity at end of stage 1.

CFA LM 8 EXAMPLE 10 — IBM TWO-STAGE DDM (END 2018)

D_{2018} = USD 6.21. g_S = 5% for one year (aligned with Value Line 2019 forecast). g_L = 2.33% thereafter (produces 8-y CAGR of 4.5%, matching forecast). r from CAPM: 2% + 0.95 × 5% = 6.75%.

1. D_{2019} = 6.21 × 1.05 = 6.5205
2. D_{2020} = 6.21 × 1.05 × 1.0233 = 6.6724
3. V_{2019} (terminal at end of 2019) = D_{2020} / (r − g_L) = 6.6724 / (0.0675 − 0.0233) = 150.96
4. V_{2018} = 6.5205 / 1.0675 + 150.96 / 1.0675 = 6.108 + 141.42 = USD 147.52

V_0 = USD 147.52 vs market USD 148.56 → approximately fairly valued

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §8 "Multistage DDMs", Example 10, pp. 354–358.

Free cash flow to equity is the capacity to pay dividends — start from net income, add non-cash, subtract net investment, add net borrowing

FCFE ≠ dividends actually paid. A firm with USD 950 FCFE paying USD 300 in dividends is retaining USD 650 — either for reinvestment or to build cash for future distributions / buybacks.

$$\text{FCFE} = \text{NI} + \text{D\&A} - \Delta\text{WC} - \text{Capex} + \text{Net borrowing}$$

Build step	USD m	Notes
Net income	1,000	After-tax earnings, bottom of P&L
+ Depreciation & amortization	+400	Non-cash expense, already in NI
− Increase in working capital (A/R, inv.)	−50	Revenue growth absorbs WC
− Capital expenditures (maintenance + growth)	−600	Fixed-capital investment
+ Net borrowing (new debt − repayments)	+200	New LT debt issued, net of amortization

VALUATION

$V_0 = \Sigma\ FCFE_t / (1+r)^t$. With a terminal assumption, a Gordon-style tail applies at the end of the explicit forecast.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §5 "FCFE Model", pp. 345–347.

The justified forward P/E falls out of the Gordon model as payout / (r − g) — which is why high-quality, high-growth, low-r firms print high P/Es

Watch the trap — payout up, retention down, so g = b × ROE falls too. Changing one input means rechecking the others ("dividend displacement of earnings").

$$P_0 / E_1 = \text{payout ratio} / (r - g) \quad \text{with} \quad g = (1 - \text{payout}) \times \text{ROE}$$

CFA LM 8 EXAMPLE 12 — NESTLÉ JUSTIFIED FWD P/E

Payout ratio (2013-16 avg) = 68%; retention b = 32%; long-run ROE = 21.5%; required return r = 9%.

1. g = 0.32 × 0.215 = 6.9%
2. r − g = 9.0% − 6.9% = 2.1%
3. P/E = 0.68 / 0.021

Justified forward P/E = 32.4×

payout / g	5.0%	6.0%	7.0%
55%	13.8	18.3	27.5
65%	16.3	21.7	32.5
70%	17.5	23.3	35.0
75%	18.8	25.0	37.5

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §9 "Multiples & Fundamentals", Example 12, pp. 358–362.

Method-of-comparables P/S ranks autos in one line — but cheap multiples demand a "why" before a "buy"

Law of one price: identical assets should sell for the same price. But firms in the same industry differ on growth, margins, leverage, cyclicality — dig into each before acting.

Automaker	P/S	Reading
Peugeot	0.28	Lowest P/S — Europe turnaround story, investors wary of margin volatility
Ford Motor	0.33	Cyclically exposed US market, legacy pension obligations weigh on equity
General Motors	0.36	Similar to Ford but with cleaner balance sheet post-2009 restructuring
Nissan Motor	0.38	Alliance-structure discount; multiple governance concerns
Honda Motor	0.46	Stable Japanese quality story; diversified (motorcycles + power products)
Daimler	0.55	Premium exposure (Mercedes-Benz) supports a higher multiple
Toyota Motor	0.80	footprint

INTERPRETATION

P/S suggests Peugeot is "cheapest" — but the discount reflects execution risk and thin historical margins. Toyota "expensive" multiple embeds decades of consistent returns. Always compare multiples with ROE / growth / risk — not in isolation.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §10 "Method of Comparables", Example 13, pp. 362–364.

EV/EBITDA compares firms at the operating level, immune to capital-structure noise — the metric of choice for cross-border and LBO work

EV captures the whole capital stack the acquirer pays for — equity + debt − cash. EBITDA is pre-D&A, pre-interest, pre-tax — the cash engine, before finance / accounting choices.

EV = Market cap + Preferred + Debt (market value) − Cash & equivalents

where: Use market value of debt when available; book value is a rough fallback. Subtract excess cash because an acquirer can use it to retire debt.

CFA LM 8 EXAMPLE 16 — CAMECO CORPORATION (FY2017)

395.8 m shares at CAD 14.11 → equity 5,585. Book debt 1,905 → market debt 1,480 (discounted using risk-free + spread curve); other liabilities 1,014. Cash 592. EBITDA 606.

1. Market cap = 395.8 × 14.11 = CAD 5,585 m
2. Market value of total debt & liabilities ≈ 1,480 + 1,014 = CAD 2,494 m

EV / EBITDA = 12.4× — benchmark against uranium peers; negative earnings years still computable (EBITDA usually positive).

WHEN EV BEATS MARKET CAP

Comparing firms with different leverage · M&A / LBO analysis · negative-earnings periods · cross-border with different tax regimes. Watch: EBITDA adds back non-cash D&A — in capital-intensive industries, that overstates cash generation.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §11 "Enterprise Value", Example 16, pp. 367–370.

Asset-based valuation sets a floor — useful for asset-heavy, intangible-light, or winding-down businesses, but weak when value sits in brands, people, or growth

Value = fair value of assets − fair value of liabilities − preferred. In practice, book-to-market adjustment is the hardest step.

WORKS — FAMILY LAUNDRY

Tangible assets dominate: the building, land, washers & dryers, inventory. Each has a verifiable market value (real-estate appraisal, used-equipment dealer).

Intangibles (location, marketing) are small. Asset-based approach yields a defensible number.

FLOOR — SECRET-RECIPE RESTAURANT

Kitchen equipment + inventory − liabilities produces a baseline.

But the "secret recipe" and the chef's reputation are intangibles that walk out with the owner.

Asset-based = floor; going-concern value requires multiple or DCF.

REVIVES — LEGACY AIRLINE

The going concern is losing money, so DCF produces a negative value.

But landing slots at congested hubs, aircraft, gate leases, and frequent-flier program are worth a lot to a competitor.

Asset-based approach is the RIGHT lens for an acquirer.

Source: CFA Institute, Curriculum Vol 5, Learning Module 8, §12 "Asset-Based Valuation", Examples 18–20, pp. 371–374.

KEY TAKEAWAYS

Deck 12 — Equity II — Analysis & Valuation

- An initial research report is built on three blocks: company (past & present), industry & competitive analysis, and forecasts — all feeding valuation and a recommendation.
- Business model = who buys × what we sell × how we make money × the competitive strategy. Decompose revenue into volume × price × mix × FX — pricing power is the decisive lever.
- Porter's Five Forces sets the long-run industry profitability ceiling. PESTLE captures the external growth-and-share tides. Rank the forces — one or two always dominate.
- Four forecast approaches: historical, base-rate convergence, management guidance, analyst discretionary. Triangulate top-down (market × share) with bottom-up (volume × price).
- A bear / base / bull scenario triangle quantifies the risks identified in Porter and PESTLE — the iPad-vs-PC case shows a USD 2.2bn FY+1 base-case revenue hit to Microsoft.
- Gordon growth DDM (Siemens, V_0 = €185.70) and two-stage DDM (IBM, V_0 = USD 147.52) are the workhorses for dividend-paying companies. FCFE replaces dividends with dividend-paying capacity.
- Justified forward P/E = payout / (r – g). Nestlé: 0.68 / (0.09 – 0.069) = 32.4× — and very sensitive to ROE and payout inputs.
- EV / EBITDA (Cameco = 12.4×) beats P/E across capital structures and negative-earnings years. Asset-based valuation sets a floor — useful when intangibles are small or the going concern is gone.